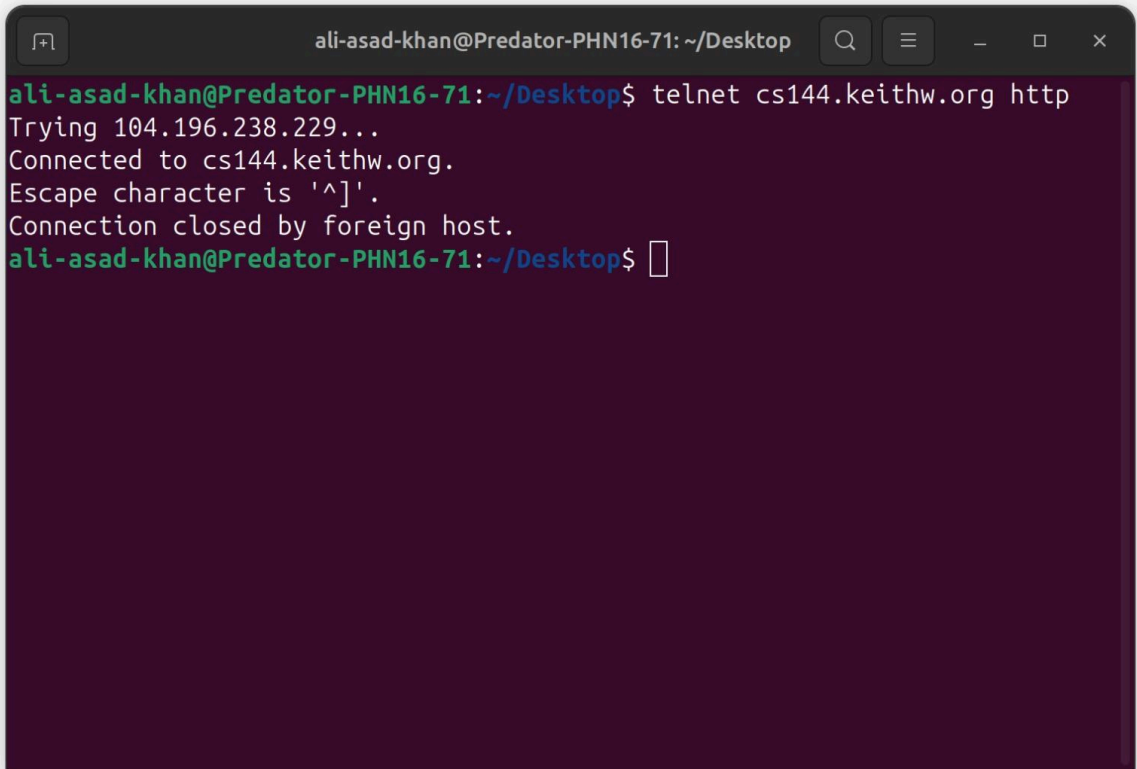


Screenshots provided for confirmation

-----2.1-----



```
ali-asad-khan@Predator-PHN16-71: ~/Desktop
ali-asad-khan@Predator-PHN16-71:~/Desktop$ telnet cs144.keithw.org http
Trying 104.196.238.229...
Connected to cs144.keithw.org.
Escape character is '^]'.
Connection closed by foreign host.
ali-asad-khan@Predator-PHN16-71:~/Desktop$
```

```
ali-asad-khan@Predator-PHN16-71:~/Desktop$ telnet cs144.keithw.org http
Trying 104.196.238.229...
Connected to cs144.keithw.org.
Escape character is '^]'.
GET /hello HTTP/1.1
Host: cs144.keithw.org
Connection: close

HTTP/1.1 200 OK
Date: Tue, 09 Sep 2025 04:10:20 GMT
Server: Apache
Last-Modified: Thu, 13 Dec 2018 15:45:29 GMT
ETag: "e-57ce93446cb64"
Accept-Ranges: bytes
Content-Length: 14
Connection: close
Content-Type: text/plain

Hello, CS144!
Connection closed by foreign host.
ali-asad-khan@Predator-PHN16-71:~/Desktop$
```

```
ali-asad-khan@Predator-PHN16-71:~/Desktop$ telnet cs144.keithw.org http
Trying 104.196.238.229...
Connected to cs144.keithw.org.
Escape character is '^]'.
GET /lab0/l230802_l232501 HTTP/1.1
Host: cs144.keithw.org
Connection: close

HTTP/1.1 200 OK
Date: Tue, 09 Sep 2025 04:23:11 GMT
Server: Apache
X-You-Said-Your-SunetID-Was: l230802_l232501
X-Your-Code-Is: 16019
Content-length: 119
Vary: Accept-Encoding
Connection: close
Content-Type: text/plain

Hello! You told us that your SUNet ID was "l230802_l232501". Please see the
HTTP headers (above) for your secret code.
Connection closed by foreign host.
ali-asad-khan@Predator-PHN16-71:~/Desktop$ █
```

```
adeenaKhan@ubuntu24: ~/Desktop
adeenaKhan@ubuntu24:~/Desktop$ netcat -v -l -p 9090
Listening on 0.0.0.0 9090
Connection received on localhost 46844
hello
testing connection from netcat window
testing connection from telnet window
^C
adeenaKhan@ubuntu24:~/Desktop$
```

```
adeenaKhan@ubuntu24: ~
adeenaKhan@ubuntu24:~$ telnet localhost 9090
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
hello
testing connection from netcat window
testing connection from telnet window
Connection closed by foreign host.
adeenaKhan@ubuntu24:~$
```

```
adeenaKhan@ubuntu24: ~/Desktop/minnow-Stanford
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$ cmake --build build
[ 10%] Built target minnow_testing_debug
[ 15%] Building CXX object src/CMakeFiles/minnow_debug.dir/byte_stream.cc.o
[ 57%] Built target util_debug
[ 68%] Built target stream_copy
[ 73%] Linking CXX static library libminnow_debug.a
[ 78%] Built target minnow_debug
[ 84%] Linking CXX executable webget
[ 89%] Linking CXX executable tcp_native
[ 94%] Built target webget
[100%] Built target tcp_native
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$ cmake --build build --target check_webget
Test project /home/adeenaKhan/Desktop/minnow-Stanford/build
  Start 1: compile with bug-checkers
1/2 Test #1: compile with bug-checkers ..... Passed    1.80 sec
  Start 2: t_webget
2/2 Test #2: t_webget ..... Passed    1.33 sec

100% tests passed, 0 tests failed out of 2

Total Test time (real) =  3.16 sec
Built target check_webget
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$
```

```
adeenaKhan@ubuntu24: ~/Desktop/minnow-Stanford
adeenaKhan@ubuntu24:~/Desktop$ cd minnow-Stanford
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$ cmake --build build --target check0
Test project /home/adeenaKhan/Desktop/minnow-Stanford/build
  Start 1: compile with bug-checkers
1/11 Test #1: compile with bug-checkers ..... Passed    15.25 sec
  Start 2: t_webget
2/11 Test #2: t_webget ..... Passed    1.34 sec
  Start 3: byte_stream_basics
3/11 Test #3: byte_stream_basics ..... Passed    0.06 sec
  Start 4: byte_stream_capacity
4/11 Test #4: byte_stream_capacity ..... Passed    0.02 sec
  Start 5: byte_stream_one_write
5/11 Test #5: byte_stream_one_write ..... Passed    0.02 sec
  Start 6: byte_stream_two_writes
6/11 Test #6: byte_stream_two_writes ..... Passed    0.02 sec
  Start 7: byte_stream_many_writes
7/11 Test #7: byte_stream_many_writes ..... Passed    0.12 sec
  Start 8: byte_stream_stress_test
8/11 Test #8: byte_stream_stress_test ..... Passed    0.04 sec
  Start 37: no_skip
9/11 Test #37: no_skip ..... Passed    0.01 sec
  Start 38: compile with optimization
10/11 Test #38: compile with optimization ..... Passed    3.33 sec
  Start 39: byte_stream_speed_test
  ByteStream throughput (pop length 4096): 2.16 Gbit/s
  ByteStream throughput (pop length 128): 1.19 Gbit/s
  ByteStream throughput (pop length 32): 0.54 Gbit/s
11/11 Test #39: byte_stream_speed_test ..... Passed    0.83 sec

100% tests passed, 0 tests failed out of 11

Total Test time (real) = 21.06 sec
Built target check0
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$
```